

Seth Baird

Motivated and ambitious full-time student at Purdue University

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EDUCATION

- **Purdue University** West Lafayette, Indiana
Anticipated Graduation May 2027
Major in Computer Engineering Technology Minor in Information Technology GPA 3.09
- **Memorial High School** Frisco, Texas
Graduated May 2023
Vocational Education: CS3: Data Structures and Algorithms, Mobile Application Development GPA 3.81

INVOLVEMENT

- **Purdue Electric Racing: Software Team (STM32 Embedded C)**
 - Developed multiple HALs (hardware abstraction layers) to clean up the codebase and make development easier.
 - Reworked the CAN communication backend to use two busses simultaneously for new high-throughput motors.
 - Utilized testing equipment such as CANable to verify the hardware, and wrote detailed technical documentation.

SKILLS AND PROJECTS

- **Computer Engineering**
 - Designed and built a miniature guitar amplifier with a CAD-drawn case and a purpose-built PCB laid out in KiCad. The six-week timeline incorporated essential design tools such as system diagrams, work breakdown structures, specific requirements, functional decomposition, and circuit simulation.
 - Modernized an analog superheterodyne FM radio built in lab with automated channel scan, presets, and a display. The implementation used I2C to communicate with the tuning DAC, and DSP to run frequency calculations.
 - Built industrial control systems hands-on in lab and programmed PLCs using the IEC 61131-3 standards.
 - Created and simulated concurrent logic circuits and HDL code for Altera FPGAs in Quartus.
- **Software Development and Information Technology**
 - Developed an android app that integrated with a car's engine computer to provide data and insights to the driver. Performance was audited and data filtering was employed early on to maintain real-time responsiveness.
 - Implemented a local maxima locator algorithm that analyzed a continuous stream of RSSI data to pick out the strongest signals. This was then coupled with quartic regression to create a feedback loop for VCO tuning.
 - Devised a bidirectional off-site backup solution using a peer-to-peer encrypted tunnel and cron for scheduling.
 - Configured a domain filtering DNS server to block advertisements and malicious sites on all network devices.

WORK EXPERIENCE

- **SMS InfoComm (Wistron Corporation)** Dallas, TX
NVIDIA Server Repair Technician May 2024 - August 2024
 - **System Repair and Troubleshooting** Diagnosed, refurbished, and built NVIDIA HGX 8-GPU units.
 - **Hardware Validation and Analysis** Used proprietary diagnostic tools to identify errors and discrepancies within motherboard subsystems such as power delivery, signal integrity, and FPGAs before executing a repair.
 - **Team Workflow Optimization** Developed a Python script to automate thousands of lines of by-hand data entry into the company's antiquated database, saving my team 80 hours of labor per week.
- **Jawa.gg** Remote
Verified Seller October 2021 - August 2023
 - **Top 10 Seller in Overall Performance** Recognized by CEO Amanda Stefan for driving Jawa's growth.
 - **Computer Sales and Commissions** Built and sold dozens of tailored desktop, laptop, and Mac computers to end users at different price points and configurations.
 - **Part Refurbishment and Sales** Repaired, tested, and refurbished parts and sold them to PC builders and end users through Jawa's marketplace.